

MASS AND KINETIC ENERGY OF THE HOMUNCULUS NEBULA AROUND η CARINAE¹

NATHAN SMITH,^{2,3} ROBERT D. GEHRZ,³ PHILIP M. HINZ,⁴ WILLIAM F. HOFFMANN,⁴ JOSEPH L. HORA,⁵
 ERIC E. MAMAJEK,⁴ AND MICHAEL R. MEYER⁴

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ABSTRACT

We examine thermal infrared images of η Car at wavelengths from 4.8 to 25 μm obtained with the MIRAC3 camera system at the Magellan Observatory, and we conclude that η Car’s circumstellar Homunculus Nebula contains much more mass ($\gtrsim 12 M_{\odot}$) and kinetic energy ($10^{49.6}–10^{50}$ ergs) than had previously been recognized. Excess far-infrared emission from a large mass of cold dust had been discovered recently by Morris et al., which they attributed to 110 K dust in a compact equatorial torus. Our images suggest, however, that roughly 10 $M_{\odot}$ or more resides instead in the rapidly expanding polar lobes of the Homunculus, known to be ejected during η Car’s “Great Eruption” in the 19th century. Combining this large mass with reliable expansion speeds for the polar lobes allows us to estimate the large kinetic energy and efficient momentum transfer associated with the eruption. Most mass resides in a cool outer layer of the lobes, with dust at ~ 140 K. This material is only revealed at 18–25 μm with high spatial resolution images that can separate adjacent dust components. Warmer dust at ~ 200 K that has been recognized for some time (responsible for previous mass estimates of 2–3 $M_{\odot}$) blankets the inner surfaces of the mostly hollow polar lobes as they are irradiated more directly by the central engine.

Key words: circumstellar matter — ISM: individual (Homunculus Nebula) — stars: individual (η Carinae)

1. INTRODUCTION

The infamous massive star η Car has been the subject of numerous investigations at infrared (IR) wavelengths since Westphal & Neugebauer (1969) first showed that it is the brightest extrasolar system object in the sky at 10 and 20 μm . This IR emission results from starlight absorbed and then re-emitted by dust in the bipolar circumstellar Homunculus Nebula, which was ejected by the star during its “Great Eruption” in the mid-19th century. The total IR luminosity is more than $4 \times 10^6 L_{\odot}$ at a distance of 2.3 kpc (Cox et al. 1995; Davidson & Humphreys 1997). Extensive mid-IR imaging and photometry of η Car has revealed numerous clues to its circumstellar structure and dust properties (Gehrz et al. 1973; Robinson, Hyland, & Thomas 1973; Joyce 1975; Aitken & Jones 1975; Hyland et al. 1979; Mitchell et al. 1983; Hackwell, Gehrz, & Grasdalen 1986; Smith et al. 1995). Recently, Smith, Gehrz, & Krautter (1998) reviewed previous thermal-IR observations of η Car and presented multiwavelength IR array images showing the spatial distribution of dust emission. From these data it was inferred that the Homunculus contains at least 2.5 $M_{\odot}$ if the gas-to-dust mass ratio is at least 100. Polomski et al. (1999) published a multiwavelength analysis with similar images and results.

However, Morris et al. (1999) presented a 2–200 μm spectrum of η Car obtained with the *Infrared Space Observatory* (ISO) that revealed far-IR emission exceeding what had been inferred from previous mid-IR data. Morris et al. attributed the far-IR excess emission to a 110 K dust component residing in a compact and very massive circumstellar torus, but Davidson & Smith (2000) showed that such a compact cool torus could not produce the observed emission. Here we attempt to ascertain the location of the dust that presumably gives rise to the important excess far-IR emission detected by Morris et al. (1999).

In a previous paper (Smith et al. 2002), we presented new high-resolution thermal-IR images of η Car obtained on 2001 August 8 with the BLINC/MIRAC3 camera system on the Baade Telescope of the Magellan Observatory. The observations and basic steps for the reduction of these data have already been described (Smith et al. 2002). That paper gave a qualitative description of the circumstellar structure, especially what seemed to be a remarkable disrupted circumstellar dust torus. The purpose of the present investigation is to present a quantitative analysis of those data in order to decipher the relative spatial distribution and mass of hot, warm, and cold dust in the Homunculus. Specifically, these data included the first high-resolution IR images at wavelengths out to 25 μm , and they are useful for revealing the spatial distribution of cool dust. The mass and location of various dust components have important implications for the kinetic energy associated with the ejecta, since their ejection age and expansion speeds are known. In § 2 we examine the dust color temperature and opacity, and in § 3 we examine the spectral energy distribution (SED) at various positions in the nebula. In § 4 we combine these clues with the previously published ISO spectrum of η Car (Morris et al. 1999) to determine the likely mass distribution in the nebula, and in § 5 we comment on the implications for the kinetic energy of the Great Eruption.

¹ Based on observations made at the Baade Telescope of the Magellan Observatory, a joint facility of the Carnegie Observatories, Harvard University, the Massachusetts Institute of Technology, the University of Arizona, and the University of Michigan.

² Center for Astrophysics and Space Astronomy, University of Colorado, 389 UCB, Boulder, CO 80309; nathans@casa.colorado.edu.

³ Department of Astronomy, University of Minnesota, 116 Church Street, SE, Minneapolis, MN 55455.

⁴ Steward Observatory, University of Arizona, 933 North Cherry Avenue, Tucson, AZ 85721-0065.

⁵ Harvard-Smithsonian Center for Astrophysics, 60 Garden Street, Cambridge, MA 02138.

2. ANALYSIS OF THE THERMAL-IR IMAGES

Figure 1 shows isophotal contour maps for the 4.8–25 μm MIRAC3 images originally presented by Smith et al. (2002). The position of the central star, represented by the plus sign in each panel, was determined by a careful spatial correlation of these images with *HST*/WFPC2 images (Morse et al. 1998); the star's position is not necessarily coincident with the strongest central emission peak in each image, shifting toward the northwest at

longer wavelengths. Of the six MIRAC3 filters used, only the 10.3 μm filter has a significant silicate contribution. Therefore, if we assume optically thin emission, we can use the thermal-IR continuum in the other filters to estimate grain color temperature and emission opacity at each position, after making corrections for the wavelength-dependent spatial resolution (see Smith et al. 2002).

Maps of dust grain color temperature estimated from images in two different pairs of wavelengths are shown in

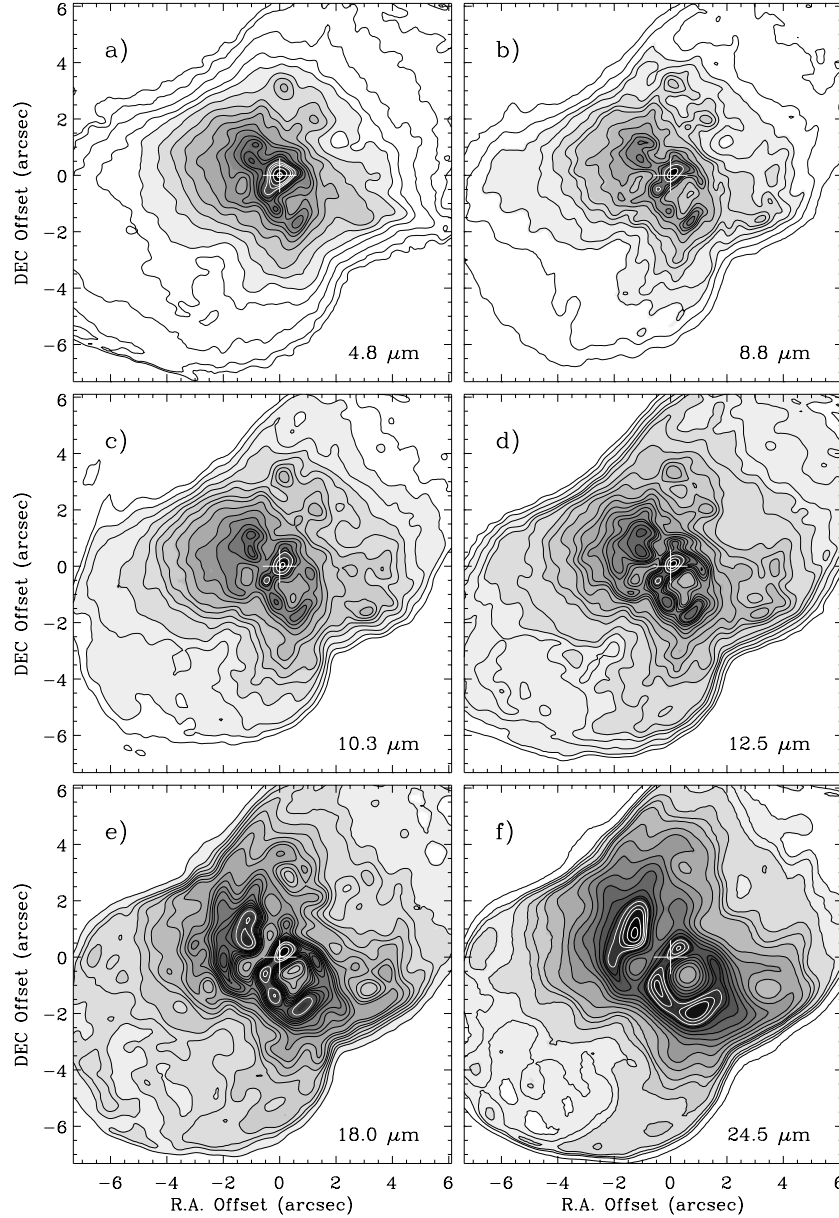


FIG. 1.—MIRAC3/Magellan images of η Car at the indicated wavelengths. Isophotal contours in Jy arcsec^{-2} are drawn at the following levels for each panel: (a) 1, 2, 4, 7, 10, 15, 22, 33, 50, 75, 100, 150, 210, 260, 350, 500, 700, 1000, 2000, and 4000 at 4.8 μm ; (b) 25, 50, 85, 125, 190, 250, 320, 500, 700, 1000, 1300, 1700, 2500, and 5000 at 8.8 μm ; (c) 65, 100, 150, 220, 300, 390, 500, 620, 850, 1150, 1500, 1900, 2400, 2700, 3000, 4500, and 7000 at 10.3 μm ; (d) 50, 70, 100, 140, 170, 210, 310, 410, 510, 620, 750, 870, 1000, 1200, 1350, 1550, 1700, 2100, 2550, 3000, 4500, and 6000 at 12.5 μm ; (e) 30, 60, 120, 200, 250, 300, 375, 450, 520, 590, 750, 870, 970, 1050, 1200, 1350, 1450, 1550, 1700, 2000, 2500, and 3200 at 18.0 μm ; (f) 30, 60, 120, 180, 220, 260, 310, 370, 420, 500, 600, 720, 840, 960, 1100, 1250, 1350, 1450, 1600, 1750, 1900, and 2050 at 24.5 μm . A plus sign marks the position of the central star in all panels.

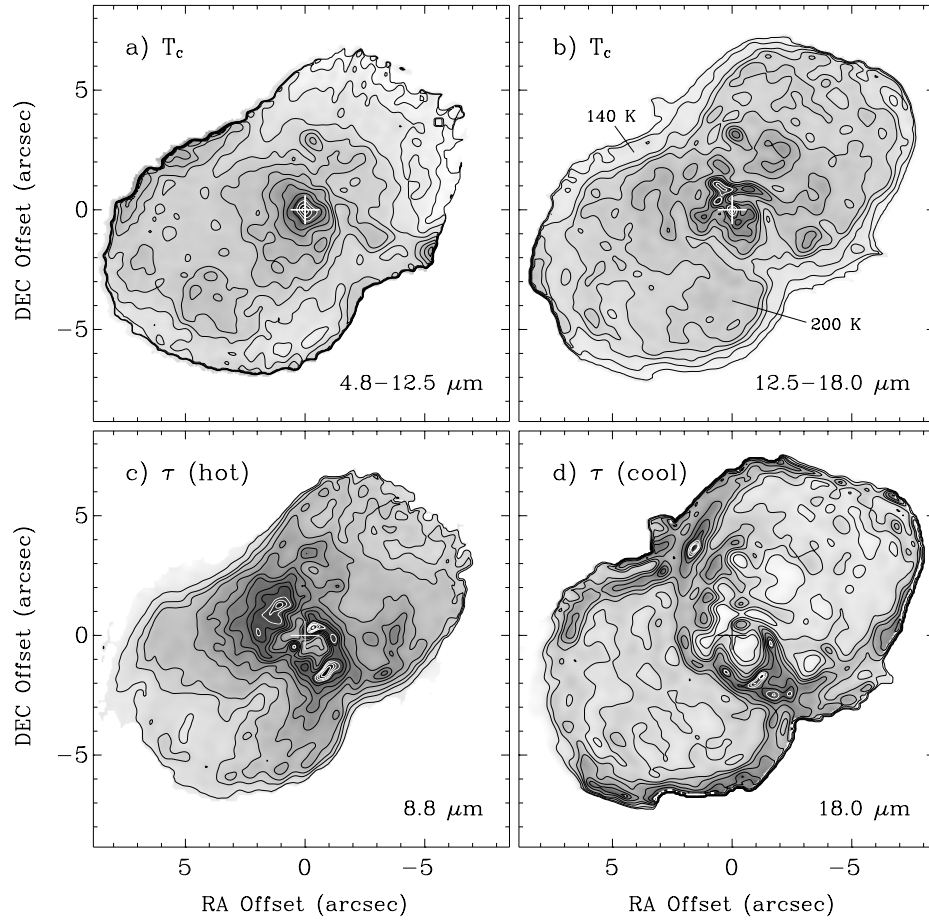


FIG. 2.—(a) Map of the observed 4.8–12.5 μm dust color temperature with contours at 220, 230, 240, 250, 260, 270, 280, 290, 310, 330, 360, 390, 420, and 450 K. Temperatures in the lobes are somewhat overestimated due to a contribution of scattered light at 4.8 μm . (b) Map of the observed 12.5–18.0 μm dust color temperature with contours at 120, 140, 160, 180, 200, 240, 280, 320, 360, 400, 450, 550, and 650 K. (c) Emission optical depth of hot dust at 8.8 μm corresponding to the color temperature map in panel (a). Contours are drawn at optical depths of 0.01, 0.02, 0.03, 0.05, 0.08, 0.11, 0.14, 0.18, 0.22, 0.27, 0.32, 0.37, and 0.44. All the emission appears to be optically thin at IR wavelengths below 10 μm . (d) Emission optical depth of cool dust at 18.0 μm corresponding to the color temperature map in panel (b). Contours of optical depth are drawn at 0.05, 0.1, 0.2, 0.3, 0.5, 0.8, 1.1, 1.4, 1.8, and 2.2; optically thick contours are drawn in white in panel (d). A plus sign marks the position of the central star in all panels.

Figures 2a and 2b. The color temperature for each pixel in these maps is given by

$$T_c = \frac{14404[(1/\lambda_2) - (1/\lambda_1)]}{\ln\{[F_\nu(\lambda_1)/F_\nu(\lambda_2)](\lambda_1/\lambda_2)^{\beta+3}\}} \text{ K}, \quad (1)$$

where $F_\nu(\lambda_1)/F_\nu(\lambda_2)$ is the flux ratio between the continuum at any two wavelengths λ_1 and λ_2 (μm),⁶ and $-\beta$ is the dust emissivity exponent. These maps were constructed assuming that at these wavelengths the grains have an emissivity proportional to λ^{-1} (i.e., $\beta = 1$). The emissivity for the grains around η Car is not known precisely, but values of β near 1 are more realistic than pure blackbody emission due to grain emission efficiencies at IR wavelengths (e.g., Van de Hulst 1957; Gehrz & Ney 1992). Smaller values of β will result in slightly higher derived color temperatures. The two maps look somewhat different because the apparent grain

temperature that dominates the emission in each wavelength range is wavelength-dependent. Information about the relative grain temperatures as a function of position in the nebula can also be gleaned from the three-color image in Figure 3a, made by coding 18.0 μm emission as red, 12.5 μm as green, and 8.8 μm as blue (i.e., hotter dust is bluer).

Color temperature maps in Figures 2a and 2b were used to derive the 8.8 and 18.0 μm emission optical depth in Figures 2c and 2d, respectively. In a given filter, emission optical depth τ is given by

$$\tau = -\ln\left[1 - \frac{F_\nu}{\Omega B_\nu(T_c)}\right], \quad (2)$$

where $B_\nu(T_c)$ is the Planck function for a given color temperature and Ω is the solid angle of a pixel. These maps reveal that some locations in the bright inner core of the Homunculus—especially the limb-brightened edges of the torus—have $\tau \gtrsim 1$. The derived values for color temperature at these same locations is therefore somewhat uncertain (and will be higher than shown in Figs. 2a and 2b), but the

⁶ Some readers may cringe at the appearance of λ and F_ν together in the same equation or figure, but this usage is convenient because of observational conventions.

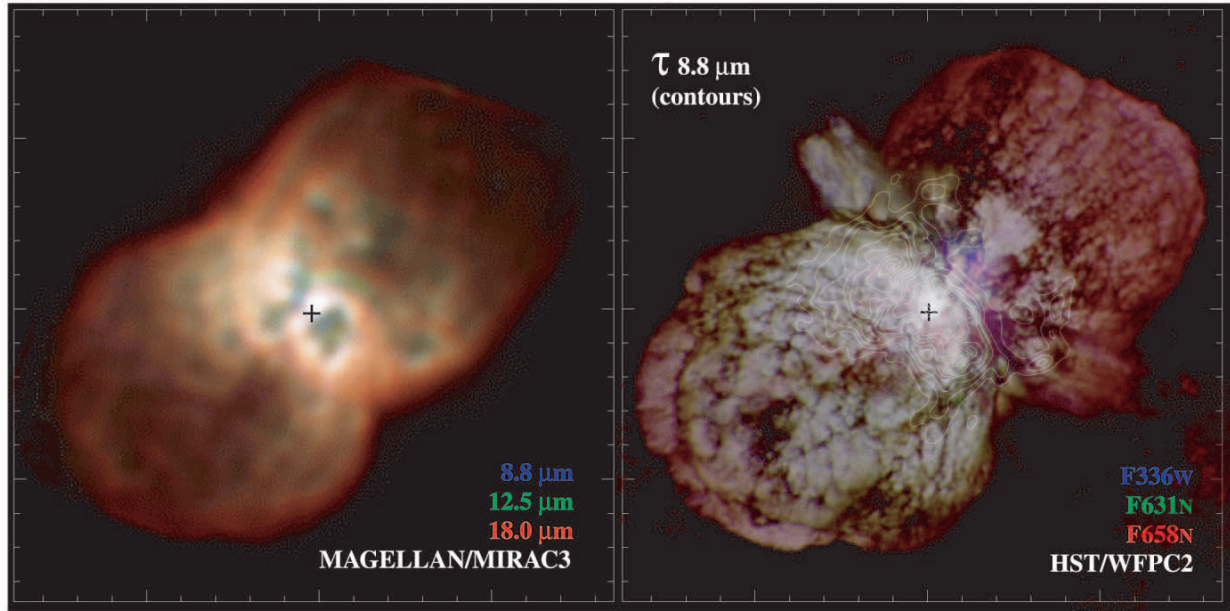


FIG. 3.—*Left*: Composite three-color thermal-IR image of η Car made from the MIRAC3/Magellan data at 8.8 (blue), 12.5 (green), and 18.0 (red) μm . *Right*: Composite three-color image made from *HST*/WFPC2 data in the F336W (blue), F631N (green), and F658N (red) filters (see Morse et al. 1998). Contours from Fig. 2c are superposed.

maps in Figure 2 still give an accurate description of the relative temperature and optical depth distribution. Contours of the 8.8 μm opacity are also superposed on the color *HST*/WFPC2 image in Figure 3b.

3. COMPONENTS OF THE IR STRUCTURE

Elsewhere, we have already given a qualitative description of the interesting morphology seen in the high-

resolution MIRAC3 images of η Car (Smith et al. 2002). In this section we concentrate on aperture photometry of selected features in the inner core and polar lobes of the Homunculus. Figure 4 shows several rectangular apertures drawn on the 12.5 μm gray-scale image; the SED for each of these apertures is shown in Figure 5. For the SED corresponding to each aperture, we have plotted blackbody continuum curves with λ^{-1} emissivity for various temperatures. These temperature components are chosen for comparison

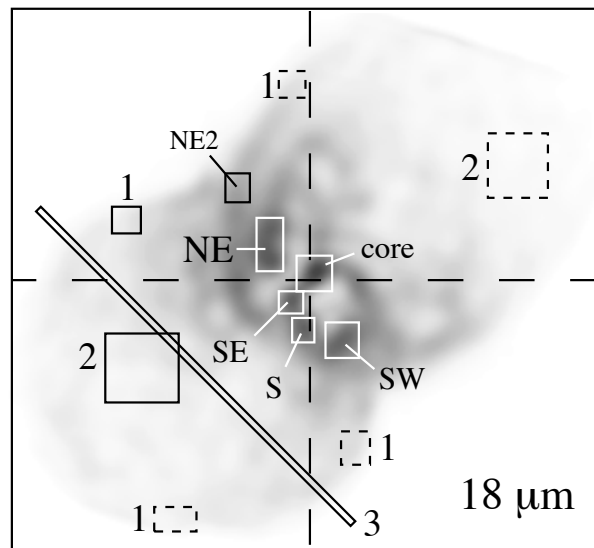


FIG. 4.—Locations and sizes of apertures used for measuring the SEDs at various positions in the nebula superposed on the 18.0 μm image; identifications correspond to SEDs in Fig. 5. The SEDs for the dotted boxes labeled 1 and 2 are not shown in Fig. 5, but they are comparable to the SEDs for the corresponding solid boxes labeled 1 or 2, respectively. The long slit labeled 3 corresponds to the spatial intensity tracings in Fig. 6.

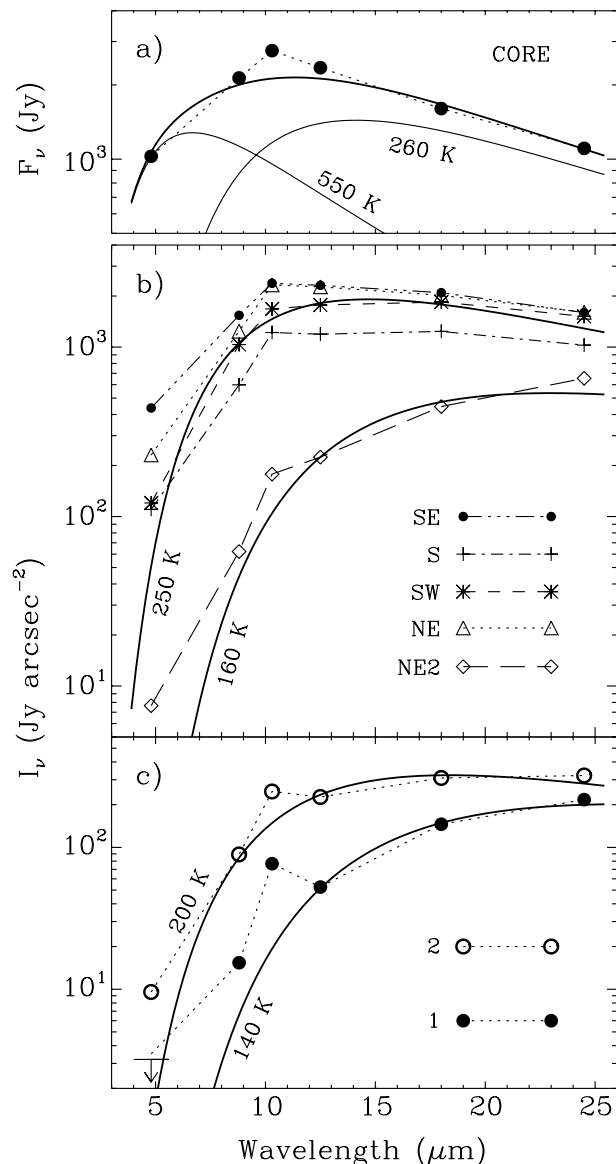


FIG. 5.—SEDs for various positions in the Homunculus (see Fig. 4). (a) Flux for the bright central core of η Car measured in the $0''.9 \times 0''.9$ aperture shown in Fig. 4. This includes the central star and the Weigelt blobs. The solid curve through the data is the sum of two blackbodies at 550 and 260 K; here and elsewhere we assume an emissivity proportional to λ^{-1} . (b) Average intensity for the apertures centered on bright spots in the circumstellar torus, labeled as SE, S, SW, NE, and NE2 in Fig. 4. Most of these have a similar SED, and a 250 K curve is drawn for comparison. The position labeled NE2 has a cooler SED, and a 160 K curve is drawn. (c) Average intensity for positions 1 and 2 in the SE lobe; position 2 is in the middle of the SE lobe and seems to be dominated by 200 K dust, while position 1 is at the limb-brightened edge and shows cooler dust at roughly 140 K. The dotted apertures in Fig. 4 labeled as 1 or 2 have very similar SEDs to the corresponding components shown here.

to represent the characteristic dust temperature that seems to dominate the energy distribution at each position, but a range of dust temperatures and grain sizes may also be present.

3.1. The Central Star and Weigelt Blobs

Figure 5a shows the observed SED for the central $0''.9 \times 0''.9$ square aperture labeled “core” in Figure 4. From high-resolution optical images (e.g., Morse et al. 1998; Weigelt & Ebersberger 1986) we know that this aperture includes the central star, as well as three nebular emission-line condensations called the “Weigelt blobs,” which are located $\sim 0''.3$ northwest of the star. A single-temperature blackbody cannot fit the SED in Figure 5a, and it is likely that there are at least two separate dust components because the source position shifts with wavelength. From images in Figure 1, we can see that at $4.8 \mu\text{m}$ the observed flux in this aperture is dominated by hot dust coincident with the bright central star, while at 18 and $24.5 \mu\text{m}$ the peak emission has migrated to the northwest direction and is therefore dominated by emission from dust in circumstellar condensations. With this in mind, we chose a two-component fit in Figure 5a with hot dust at 550 K representing dust close to the star and warm dust at 260 K in the Weigelt blobs. The blackbody equilibrium temperature for grains around η Car is given by

$$T_{\text{BB}} \approx 13,100 R_{\text{AU}}^{-1/2} \text{ K}, \quad (3)$$

where R_{AU} is the distance from the central engine in AU. T_{BB} would be about 400 K at the distance of the Weigelt blobs, which is roughly $0''.3$, or 900 AU, if they reside in the equator (see Davidson & Humphreys 1997). Therefore, it is likely that their dust is partly shielded from the star’s UV and optical emission. The hot $\gtrsim 550$ K component is consistent with previous results of imaging photometry (Hackwell et al. 1986; Smith et al. 1995, 1998) if differences in spatial resolution are considered. This hot dust located $\lesssim 0''.15$ from the star may have formed during mass-loss episodes over the past 5–20 yr; the $\sim 10 \mu\text{m}$ flux may be variable and should be monitored.

The 260 K component fit in Figure 5a allows us to make a crude estimate of the dust mass contained in the Weigelt blobs for the first time. With a known distance $D \approx 2300$ pc, and with an assumed grain density $\rho \approx 3 \text{ g cm}^{-3}$ and radius $a \approx 1 \mu\text{m}$ (see Smith et al. 1998), the mass of dust needed to emit an observed flux F_λ at a given wavelength λ is

$$M_{\text{dust}} = \frac{4D^2 \rho (\lambda F_\lambda)}{3(\lambda Q_e/a) B_\lambda(T)}, \quad (4)$$

where $(\lambda Q_e/a)$ is a quantity that describes the grain emission efficiency Q_e (see Draine & Lee 1984).⁷ The 260 K component in Figure 5a indicates a dust mass of roughly $3 \times 10^{-4} M_\odot$ contained in the Weigelt blobs.⁸ If the gas-to-dust mass ratio is $\gtrsim 100$, then the Weigelt blobs probably contain at least $0.03 M_\odot$ of material.

This central aperture also shows a significant $10 \mu\text{m}$ silicate emission feature in Figure 5a. The position of the central emission peak in the $10.3 \mu\text{m}$ filter in Figure 1c is intermediate between adjacent filters, so the silicate dust appears to be distributed evenly between the two tempera-

⁷ This assumes optically thin emission, which seems reasonable for the Weigelt blobs, but may not apply for some of the condensations in the torus (see Fig. 2d). If high optical depths are important in these other condensations, then eq. (4) gives an underestimate of the mass.

⁸ Thus, each of the three Weigelt blobs contains roughly 1 Jupiter’s worth of condensibles.

ture components. This suggests that silicate grains continuously form in η Car's wind or during recurring mass ejection episodes.

3.2. Clumps in the Dust Torus

Figure 5b shows the observed SEDs for several features that may be clumps in a disrupted equatorial dust torus (see Smith et al. 1995, 2002). All the bright condensations in the torus have energy distributions that are roughly consistent with dust near 250 K, much like the Weigelt blobs discussed above. Since they have fluxes comparable to the Weigelt blobs, it is likely that each of these bright clumps contains a few percent of a solar mass (again, assuming a gas-to-dust mass ratio of about 100).

One position labeled “NE2” is located farther out along the minor axis of the nebula and appears to be seen in projection in the equatorial plane. Given its low apparent temperature around 160 K, it may be outside the warm torus in equatorial dust that is shielded from the central engine. Alternatively, emission at this position may simply be dominated by the limb-brightened edges of the polar lobes. Note that NE2 is roughly the position where we expect the edges of the southeast and northwest polar lobes to overlap. This may provide additional support for the conjecture below that cool dust resides in the walls of the polar lobes.

3.3. Warm and Cool Dust in the Polar Lobes

Figure 5c shows that the observed SED at the edges of the polar lobes (position 1) has a characteristic temperature around 140 K and is therefore dominated by cooler dust than the SED seen in the middle of the lobes (position 2). Warm ~ 200 K dust had already been inferred from previous mid-IR imaging and photometry at 2–12 μm (e.g., Smith et al. 1998; Cox et al. 1995; Hackwell et al. 1986), but the cooler dust in the polar lobes is only revealed with photometric data at longer wavelengths (18–25 μm) and with high spatial resolution needed to separate adjacent dust layers. These two temperature components at 140 and 200 K are seen clearly in the 12.5–18 μm color temperature map in Figure 2b. In Figure 5c the SED at position 1 appears to have the strongest silicate feature compared to all other observed positions.

Figure 6 shows tracings of the flux, optical depth, and color temperature across the middle of the southeast polar lobe (Fig. 4, position 3). The apparent diameter of the lobes appears to be a function of wavelength, with a slightly larger size at longer IR wavelengths (Fig. 6a). This is also apparent in Figure 1; the effect is more dramatic than would be expected from differences in spatial resolution. For example, if the spatial resolution of the 12.5 μm image is degraded, a similar tracing to that in Figure 6a does not match the 18 μm profile. Figure 6b indicates clearly that the highest column density of cool dust emitting at 18 μm is seen exterior to the high column densities of warmer dust dominating at 8.8 μm . Finally, Figure 6c confirms that these column density peaks at the edges of the polar lobes are spatially coincident with the 140 K dust seen in the color temperature map, and that the warmer 200 K dust fills the space between the edges of the lobes.

All these facts suggest that cool 140 K dust emitting preferentially at long wavelengths resides just outside the warmer 200 K dust, as depicted in Figure 7. The warmer 200 K dust seems to blanket the inner surfaces of the polar lobes

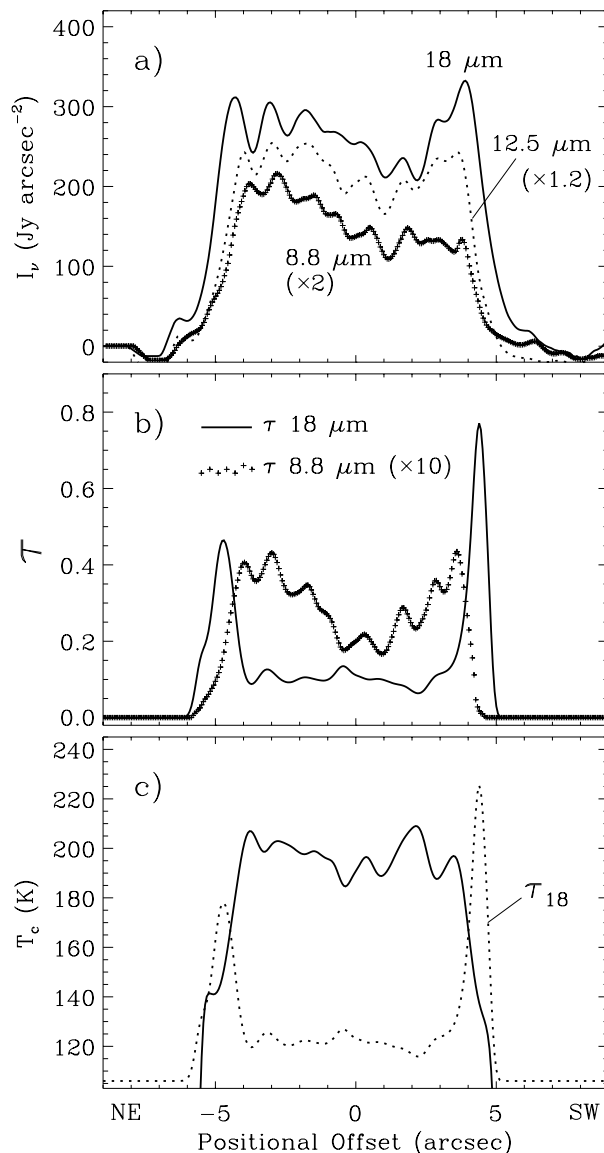


FIG. 6.—Tracings of various quantities across the slit labeled 3 in Fig. 4. (a) Specific intensity of 8.8, 12.5, and 18 μm emission; (b) 8.8 and 18 μm emission optical depth from Figs. 2c and 2d; (c) 12.5–18 μm color temperature from Fig. 2b (with 18 μm optical depth scaled arbitrarily and superposed).

as they are irradiated by the central star; the lack of strong “limb brightening” in the 8.8 μm optical depth map suggests that this warm 200 K dust may partially fill the outer regions of the mostly hollow cavity inside the lobes. The sharp temperature gradient from ~ 200 to ~ 140 K in the walls of the polar lobes over distances less than 2000 AU suggests that dust in the outer parts of the lobes is shielded—the lobes are probably optically thick to the direct UV and optical radiation that heats the grains.⁹ This

⁹ Alternatively, one may interpret the temperature gradient in the lobe walls as being due to more efficient radiative cooling from the outer surfaces down to a skin depth, also suggesting that they are optically thick.

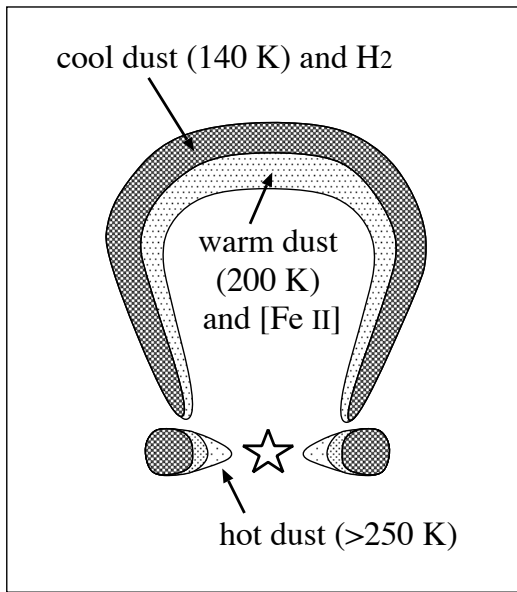


FIG. 7.—Cartoon representing the stratification of dust temperature in the Homunculus lobes (see § 3.3).

explains why Smith et al. (1998) found that, when seen in thermal emission at wavelengths near $12\ \mu\text{m}$, the polar lobes appear somewhat smaller than in scattered light at optical wavelengths. Namely, wavelengths near $12\ \mu\text{m}$ are dominated by emission from warmer 200 K dust at the inside surfaces of the polar lobes, while optical light is scattered by both the warm and cool dust. Smith et al. (1998) also found that the dust mass in the polar lobes derived from scattered continuum light at $2\ \mu\text{m}$ was much larger than that measured with thermal emission at $10\ \mu\text{m}$ —this is relevant to the results below.

The dust temperature structure for the polar lobes depicted in Figure 7 also seems to conform to the presumed structure inferred independently from near-IR spectro-

scopy: Smith (2002) found that a cool molecular shell seen in emission from $\text{H}_2\ 1-0\ S(1)$ traces the bipolar structure in the outer walls of the Homunculus lobes, while warm ionized atomic gas seen in $[\text{Fe II}]\ \lambda 16435$ resides immediately interior to the molecular gas. The cool molecular component may be nearly invisible in other molecular tracers; for instance, if high-resolution CO maps become available in the future, it is not clear that they should detect this material, since $\eta\ \text{Car}$'s ejecta may be severely depleted of C and O.

4. A LARGE MASS OF COOL DUST IN THE POLAR LOBES AND THE FAR-IR EXCESS DETECTED BY *ISO*

The discovery described in the previous section—that there is a significant component of cool shielded dust in the outer parts of the bipolar lobes—may have important consequences for the interpretation of $\eta\ \text{Car}$'s 2–200 μm energy distribution revealed by *ISO*. Morris et al. (1999) presented the *ISO* spectrum of $\eta\ \text{Car}$ (a tracing is shown in Fig. 8) and discovered that the observed far-IR flux exceeded that expected from the $\sim 2\ M_\odot$ of $\sim 200\ \text{K}$ dust in the polar lobes that had been identified earlier (Cox et al. 1995; Smith et al. 1998). Morris et al. fitted the *ISO* spectrum with three temperature components at ≥ 400 , 190, and 110 K (see Fig. 8a) and suggested that the previously unrecognized 110 K dust component resided in a compact equatorial torus containing $\sim 15\ M_\odot$ of gas and dust. However, mid-IR images (Hackwell et al. 1986; Smith et al. 1995, 1998; Polonski et al. 1999) revealed that the bright features that Morris et al. associated with the 110 K torus are warmer (250–350 K), and Davidson & Smith (2000) argued that dust with a brightness temperature of 110 K could not produce the required luminosity and would need to be distributed over a much larger area on the sky than was occupied by the putative torus. Thus, the location of the cool dust giving rise to the far-IR excess emission discovered by Morris et al. (1999) has remained unknown. This is an urgent problem because, as Morris et al. pointed out, the cool dust may be associated

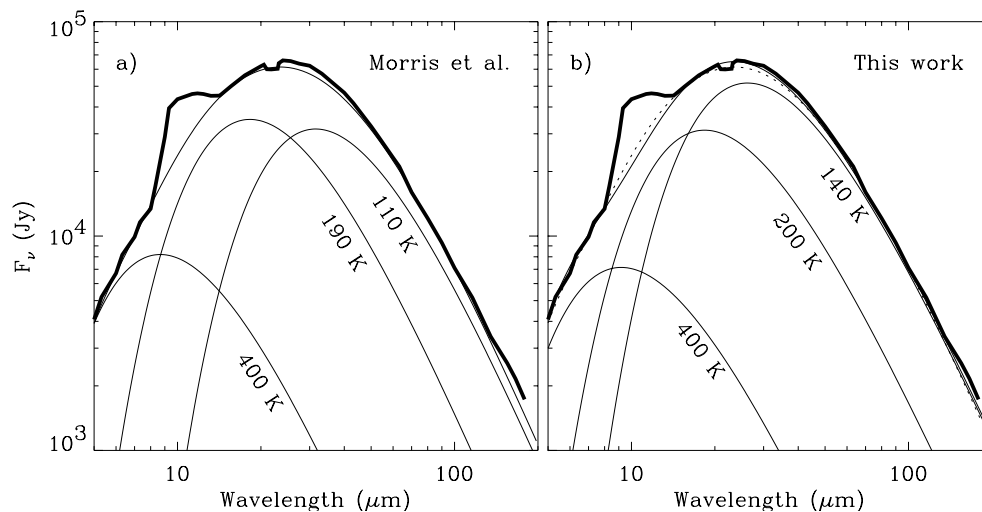


FIG. 8.—Tracing of the *ISO* spectrum of $\eta\ \text{Car}$ presented by Morris et al. (1999). Morris et al.'s fit is shown in panel (a), and also in panel (b) with the dashed line for comparison with an alternative fit using a λ^{-1} emissivity and three components at 400, 200, and 140 K.

with a majority of the mass in the Homunculus. Even at the shortest IR wavelengths, the smallest *ISO* aperture included emission from nearly the entire Homunculus, and the aperture at the longest far-IR wavelengths was much larger—more than 1' in diameter.

The *ISO* spectrum of η Car shown in Figure 8 is a smooth continuous energy distribution (except for the 10 μ m silicate feature), so Morris et al.'s fit to the continuum is not necessarily unique. The Homunculus contains a range of dust temperatures, and other satisfactory fits to the spectrum can be achieved by adjusting the number of components, as well as their temperatures and emissivities. One alternate possibility is shown in Figure 8b, which is composed of three temperature components at 400, 200, and 140 K with a λ^{-1} emissivity (Morris et al. 1999 used a $\lambda^{-1.22}$ emissivity). Again, ours is not a unique fit either.¹⁰ It is interesting, however, because it contains the ≥ 400 and 200 K temperature components that were previously recognized, as well as a significant contribution from dust at ~ 140 K. Thus, it seems likely that the 140 K dust seen at the outer edges of the polar lobes in our MIRAC3/Magellan images (see § 3.3) may make a significant contribution to the total IR luminosity and dust mass of the Homunculus. The 140 K component in our fit makes a stronger contribution in 18 and 25 μ m images than Morris et al.'s 110 K component.

The dust mass can be calculated for each temperature component from equation (4) with the same assumptions for grain properties given there; Table 1 lists these dust masses as well as the IR luminosity for each component and the total of all three. The total dust mass is relatively insensitive to the chosen temperature components; our total mass of 12.5 $M_{\odot}$ (see below) is consistent with the 10–15 $M_{\odot}$ found by Morris et al. (1999), who chose very different temperature components (Fig. 8), and is robust to the extent that we can trust the far-IR fluxes recorded by *ISO*. Assumed gas-to-dust mass ratios are far more uncertain. The results for the 400 and 200 K components are unsurprising in light of previous work, and the 140 K component accounts for the extra mass and excess far-IR flux in Morris et al.'s 110 K component. Because of various clues described above, we attribute the large mass of cooler dust mostly to

the polar lobes, instead of a massive equatorial torus. This result requires that we reassess some of the basic parameters associated with η Car's Homunculus and the Great Eruption.

5. IMPLICATIONS FOR THE GREAT ERUPTION

The analysis above indicates that the Homunculus contains roughly 0.125 $M_{\odot}$ of dust. With a conservative estimate of the gas-to-dust mass ratio of ~ 100 (it could be a factor of 2 higher), we estimate that the total mass contained in the Homunculus is at least 12.5 $M_{\odot}$. Some of this mass may be located in the putative equatorial torus. We have shown that the bright clumps in the torus are warm, and they contribute a small fraction of the total mass, but some cooler material may still be shielded and hidden in the equatorial debris (as depicted in Fig. 7). Guided by the optical depth maps in Figures 2c and 2d, it may be possible that 10%–20% of the total mass in the Homunculus may be hidden in the equatorial torus or in the low-latitude regions of the side walls of the polar lobes (as much as 2.5 $M_{\odot}$). Therefore, it is likely that at least 10 $M_{\odot}$ of material resides in the thick caps of the polar lobes.

This previously unrecognized large mass of material in the polar lobes places valuable constraints on the physics of the Great Eruption, because we know from proper motion measurements (Morse et al. 2001; Smith & Gehrz 1998; Currie et al. 1996) that the polar lobes were ejected during that event. There are important consequences for two fundamental parameters of the eruption:

1. *The mass-loss rate.*—If the Great Eruption lasted about 20 yr, then the time-averaged $\dot{M}$ during the event was $\geq 0.5 M_{\odot} \text{ yr}^{-1}$ (it could have been as high as 1 $M_{\odot} \text{ yr}^{-1}$ since our assumed gas-to-dust ratio is conservative). This is higher than the value of $\sim 0.1 M_{\odot} \text{ yr}^{-1}$ that is usually quoted (Davidson & Humphreys 1997). Most of the mass-loss was directed toward the poles, which poses an interesting theoretical question beyond the scope of this paper. *Caveat.*—The mass-loss rate at some times during the eruption could have been even higher than 1 $M_{\odot} \text{ yr}^{-1}$, since some proper motion measurements imply a very narrow range in ejection dates (Morse et al. 2001).

2. *The kinetic energy in the ejecta.*—The polar lobes are expanding at roughly 650 km s⁻¹, given their projected size in images and the age of the Homunculus. With a large mass of $\geq 10 M_{\odot}$, the polar lobes also carry a great deal of kinetic energy. The mechanical energy imparted to the ejecta during the Great Eruption was therefore on the order of $10^{49.6}$ – 10^{50} ergs, much higher than previously assumed. The mechanical energy in the expanding Homunculus is greater than the total amount of luminous energy presumed to be radiated by η Car during the 20 yr duration of the Great Eruption event, which was $\sim 10^{49.5}$ ergs (Davidson & Humphreys 1997; assuming that there was a negligible bolometric correction during the event).

This near equipartition between luminous and kinetic energy may be critical for understanding the acceleration of the ejecta. During the Great Eruption, η Car probably exceeded its classical Eddington luminosity by a factor of several, so one would expect radiation to play a key role in the mass loss and the structure of the super-Eddington atmosphere (see Shaviv 2000). When radiation is suspected to

¹⁰ The fit can be improved by adding another component at ~ 250 K, as one might expect from the SEDs shown in Fig. 5. However, the three-component fit shown in Fig. 8b is as satisfactory as that in Fig. 8a, and is especially significant because it was performed as one of several tests before the MIRAC3/Magellan images were obtained. In any case, adding additional temperature components to fit the IR emission is not expected to significantly alter the total mass estimate; differences between model fits are less than the uncertainty in the absolute flux calibration of the data (10%–20%).

TABLE 1
COMPONENTS FITTED TO THE *ISO* SPECTRUM

Component	$M_{\text{dust}}/M_{\odot}$	$L/L_{\odot}$	Location
400 K	2×10^{-4}	5.5×10^5	Core
200 K	0.015	1.2×10^6	Lobes, torus
140 K	0.11	1.4×10^6	Lobes
Silicate ^a	...	1.2×10^6	Mixed
Total (IR)	0.125	4.3×10^6	...

^a Silicate emission is distributed among dust at various temperatures.

drive mass loss, one often wishes to know the ratio ζ of the wind momentum to the photon momentum (sometimes called the “performance number,” usually denoted by “ η ,” which we avoid here for obvious reasons) given by

$$\zeta = \frac{\dot{M}v}{L/c}, \quad (5)$$

where $L \approx 5 \times 10^{40}$ ergs s⁻¹ is the time-averaged luminosity during the eruption for the total luminous output noted above. For η Car’s present-day wind (see Hillier et al. 2001) we find a value of $\zeta \approx 5$, which is typical compared to powerful radiatively driven winds of Wolf-Rayet stars (Abbott & Conti 1987)—but for conditions during η Car’s Great Eruption we find $\zeta > 10^3$. Performance numbers as high as 10 can result from purely line-driven winds (Lucy & Abbott 1993; Springmann & Puls 1998), but such extremely efficient momentum transfer with $\zeta = 10^3$ is unlikely, even permitting multiple scattering in an optically thick continuum-driven wind. Perhaps some of η Car’s luminous output during the eruption is unaccounted for, but it is hard to accept that it has been underestimated by a factor of 100. Thus, rather than a transfer of momentum by radiation pressure in a wind, the high value of ζ implies a more direct transfer of energy in an *explosion* (or perhaps pulsation-driven mass loss). For comparison, performance numbers near 10^3 are typical for Type II supernovae. (Zwicky’s 1964

description of the Great Eruption as a “Type V supernova” may have been somewhat appropriate after all, even though η Car survived its outburst.) In any case, the new estimated mass and kinetic energy of the Homunculus indicate that the Great Eruption was more powerful than is usually assumed. In fact, the energy quoted here is an underestimate, since we have not included kinetic energy of additional fast-moving ejecta outside the Homunculus, some of which originated in the same event. It is intriguing that the underlying cause of such a powerful event remains unknown, despite detailed observational study and more recent examples of similar extragalactic eruptions.

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